Savitribai Phule Pune University
Department of Technology
B.Sc. BlockChain Technology
F. Y. B.Sc. BlockChain Technology Sem II Practical Examination 2024-25
Subject: BBT – 205 Java Programming Lab

Time: 3 Hours Max Marks: 100

Q1. Write the programs for the following

a. Write a smart contract with a constructor that initializes a state variable owner and a mapping userValues . Create a modifier to restrict a public function setUserValue to only the owner, which updates userValues for a given address. Include a view function to read a user's value and a pure function to calculate the square of a local variable. [15]

b. Create a sales performance dashboard to visualize total sales, profit, and quantity sold by product and region. Include filters for time periods and product categories. [25]

Q2. Write a 'Java' program for the following:

a. Write a Java program to sort an array in ascending order. [15]

b. Write a Java program using File Handling to copy contents of one file to another.

[25]

Q3. Viva. [10]

Q4. Journal. [10]

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Q1. Write the program for the following:

a. Create a smart contract with a state variable balance (in wei) and mapping userBalances . Use a constructor to set an initial balance. Write a public function with a modifier to check if msg.sender is a specific address, allowing it to add ether to userBalances. Include a view function to return a user's balance and a pure function to convert a local variable from ether to wei. [15]

b. Create a customer demographics dashboard to analyze customer data, including age, gender, and location. Include slicers to filter by regions and age groups. [25]

Q2. Write a 'Java' program for the following:

a. Write a Java program to check whether a string is a palindrome or not. [15]

b. **Write a Java program using Exception Handling** to accept a number from the user. If the number is zero, throw a user-defined exception "Number is Zero, otherwise check whether the number is a palindrome or not. [25]

Q3. Viva. [10]

Q4. Journal. [10]

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Q1. Write a program for the following:

a. Write a smart contract with a state variable name and a mapping userNames . Use a constructor to initialize name. Create a modifier to ensure a public function updateUserName only executes if the new name length is greater than 3. Include a view function to read a user's name and a pure function to concatenate two local strings. [15]

b. **Create an inventory management dashboard** to track stock levels for products. Display out-of-stock and low-stock items, along with trends in product availability over time.

[25]

Q2. Write a 'Java' program for the following:

a. To check whether the given number is palindrome or not. [15]
b. **Write a Java program using inheritance** to create a class Book with attributes title and author, and a subclass LibraryBook with shelf number and availability. Display full book details. [25]

Q3. Viva. [10]

Q4. Journal. [10]

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Q1. Write a program for the following:

a. Develop a smart contract with a state variable counter and a mapping userCounters . Initialize the counter in a constructor. Write a public function incrementUserCounter restricted by a modifier that checks if msg.value is at least 1 ether. Include a view function to get a user's counter and a pure function to check if a local variable is even. [15]

b. **Create a financial performance dashboard** to compare actual spending against budget for various departments. Use bar charts and KPIs to track overages or savings.

[25]

Q2. Write a 'Java' program for the following:

a. **Write a Java program** to accept an array of n integers and display the smallest and largest elements. [15]

b. **Write a Java program using method and Exception Handling** to accept a number from the user. If the number is greater than 100, throw a user-defined exception "Number is out of Range", otherwise calculate and display the sum of digits of that number. [25]

Q3. Viva. [10]

Q4. Journal. [10]

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Q1. Write a program for the following:

a. Create a smart contract with a state variable isActive and a mapping userStatus Set isActive to true in the constructor. Write a public function toggleUserStatus with a modifier that requires block.timestamp to be greater than a specific value. Include a view function to read a user's status and a pure function to return the opposite of a local boolean variable. [15]

b. **Create a product performance dashboard** to analyze sales, profit, and units sold for different products. Include a pie chart to show sales distribution by category. [25]

Q2. Write a 'Java' program for the following:

a. To check whether the number given is perfect or not. [15]
b. **Write a Java program using static method and custom exception** to accept a number from the user. If the number is negative, throw a custom exception "Negative Number Not Allowed", otherwise calculate the factorial of the number. [25]

Q3. Viva. [10]

Q4. Journal. [10]

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Q1. Write a program for the following:

a. Write a smart contract with a state variable price (in wei) and a mapping userPrices . Initialize price in a constructor. Create a public function setUserPrice with a modifier that ensures the new price is non-zero. Include a view function to get a user's price in ether and a pure function to multiply two local uint variables. [15]

b. **Create a revenue and profit dashboard** to track monthly revenue and profit over time, using line charts and year-over-year comparison.

[25]

Q2. Write a 'Java' program for the following:

a. To check whether the number given is perfect or not. [15]

b. **Write a Java program using user-defined exception** to check whether a number is even or odd. If the number is negative, throw an exception "Negative Number Not Allowed". [25]

Q3. Viva. [10]

Q4. Journal. [10]

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Q1. Write a program for the following:

a. Develop a smart contract with a state variable totalSupply and a mapping userSupplies . Set totalSupply in a constructor. Write a public function increaseUserSupply restricted by a modifier that checks if the caller is the contract deployer is owner. Include a view function to read a user's supply and a pure function to divide a local variable by 2. [15]

b. **Create a customer churn dashboard** to track customer retention and churn rates over different periods. Include filters for customer segments and a line chart for trend analysis [25]

Q2. Write a 'Java' program for the following:

a. To calculate the sum of First and Last digit of a number. [15]

b. **Write a Java program using user-defined exception** to accept age from the user. If the age is less than 0 or greater than 120, throw an exception "Invalid Age", otherwise display a suitable message. [25]

Q3. Viva. [10]

Q4. Journal. [10]

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Q1. Write a program for the following:

a. Create a smart contract with a state variable userCount and a mapping userRegistrations Initialize userCount to 0 in a constructor. Write a public function registerUser with a modifier that requires msg.value to be exactly 0.5 ether and updates userRegistrations. Increment userCount. Include a view function to check if a user is registered and a pure function to check if a local variable is positive. [15]

b. **Create a regional sales dashboard** to visualize sales by region. Use a map to show geographic distribution and a bar chart for regional comparisons. [25]

Q2. Write a 'Java' program for the following:

a. To check whether given number is prime or not. [15]

b. **Write a Java program using Exception Handling** to accept a number from the user. If the number is negative, throw a custom exception "Negative Not Allowed", otherwise calculate the sum of all even digits in the number. [25]

Q3. Viva. [10]

Q4. Journal. [10]

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Q1. Write a program for the following:

a. Write a smart contract with a state variable message and a mapping userMessages . Set message in a constructor. Create a public function updateUserMessage with a modifier that ensures the new message is not empty. Include a view function to read a user's message and a pure function to calculate the length of a local string. [15]

b. **Create a employee performance dashboard** to track key metrics such as sales per employee, attendance, and productivity. Use bar charts for comparison. [25]

Q2. Write a 'Java' program for the following:

a. To calculate factorial of a number (Use function). [15]
b. **Write a Java program using File Handling** to read and display the contents of a text file. [25]

Q3. Viva. [10]

Q4. Journal. [10]

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Q1. Write a program for the following:

a. Develop a smart contract with a state variable locked and a mapping userLocks Initialize locked to false in a constructor. Write a public function lockUser with a modifier that checks if the user is not already locked. Include a view function to check a user's lock status and a pure function to return the bitwise AND of two local uint variables. [15]

b. **Create a product sales trend dashboard** to visualize sales trends for products over different periods (e.g., months, quarters). Include filters for different time ranges. [25]

Q2. Write a 'Java' program for the following:

a. To swap the values of two variables (Without using third variable). [15]
b. **Write a Java program using interface** Greetable having method greet(String name), and implement it in classes MorningGreeting and EveningGreeting to display suitable greetings [25]

Q3. Viva. [10]

Q4. Journal. [10]

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Q1. Write a program for the following:

a. Create a smart contract with a state variable funds (in wei) and a mapping userFunds . Set an initial value for funds in a constructor. Write a public function addUserFunds with a modifier that requires msg.value to be at least 1 ether. Include a view function to convert a user's funds to ether and a pure function to find the maximum of two local uint variables. [15]

b. **Create a yearly budget vs. actual spending dashboard** to compare budgeted values to actual expenses across different departments. Use bar charts and KPI indicators to visualize the data. [25]

Q2. Write a 'Java' program for the following:

a. **Write a Java program** to check whether a number is positive, negative, or zero. [15]

b. **Write a Java program using inheritance** to create a class Employee with employee ID and name, and a subclass Salary to compute and display total salary using basic pay and HRA.

[25]

Q3. Viva. [10]

Q4. Journal. [10]

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Q1. Write a program for the following:

a. Write a smart contract with a state variable ownerName and a mapping userOwnerNames. Initialize ownerName in a constructor. Create a public function changeUserOwner with a modifier that restricts access to msg.sender matching the original owner. Include a view function to read a user's owner name and a pure function to reverse a local string (simplified as length check). [15]

b. **Create a profit margin dashboard** to visualize profit margins by product and category. Use pie charts and bar charts to display the distribution of profits. [25]

Q2. Write a 'Java' program for the following:

a. **Write a Java program** to print the Fibonacci series up to n terms. [15]

b. Design a frame using Swing to perform addition of two numbers [25]

Q3. Viva. [10]

Q4. Journal. [10]